

Matthew J. Cockayne

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RESEARCH INTERESTS

Safe and responsible AI, ensuring AI systems are interpretable, robust, and equitable. Developing methods for bias detection and mitigation, systematic robustness evaluation under distribution shift, and interpretable-by-design architectures that enable human oversight and intervention. Applied through medical imaging, where deployment failures carry direct real-world harm, producing domain-agnostic frameworks transferable to broader safe and responsible AI challenges.

EDUCATION

PhD in Computer Science

2023-2026 (Expected)

Keele University, Keele, UK

Thesis: Safe and Responsible AI for Healthcare: Robustness, Interpretability, and Fairness in Skin Lesion Recognition

Funding: Competition-funded Department Studentship

Published 2 first-author papers (Pattern Analysis and Applications journal, AIIH 2025 conference oral presentation). Two additional first-author papers submitted to MICCAI 2026 and Image and Vision Computing (Elsevier). Co-author on Discover Computing journal paper under review. Developed open-source toolkits MedXAI and SkinCBM for reproducible research. PGR Representative for 3 consecutive years.

MSc in Artificial Intelligence and Data Science, Distinction

2021-2022

Keele University, Keele, UK

Data Analyst Internship: Developed explainable AI methods for CV-based sentiment analysis in recruitment. Turing Network Development Research Associate: Organised networking events and hackathons.

BSc in Physics, First Class Honours

2018-2021

Keele University, Keele, UK

RESEARCH EXPERIENCE

PhD Researcher - Responsible & Trustworthy AI

2023-Present

Keele University, School of Computer Science and Mathematics

- Designed robustness evaluation frameworks for transformer-based architectures, systematically benchmarking model behaviour under 15+ real-world corruption types (noise, blur, compression) and quantifying failure modes through uncertainty estimation; applied to multi-modal skin cancer detection achieving state-of-the-art performance while identifying deployment risks
Code: github.com/xrai keele/DermFormer
- Developed interpretable-by-design architectures (concept bottleneck models) enabling human-in-the-loop oversight. Users can inspect, validate, and correct intermediate reasoning steps, ensuring AI decisions remain auditable and aligned with human intent
- Conducted information-theoretic analysis of model completeness and concept synergy, revealing when models rely on spurious feature correlations versus genuine causal reasoning; implemented sparse polynomial classifiers preserving full interpretability of decision boundaries
- Built fairness-aware curriculum learning reducing demographic performance disparities by 44% and improving worst-group outcomes by 63%, demonstrating that safety-oriented training objectives can improve rather than constrain overall system performance
- Evaluated foundation model (SAM) reliability in zero-shot transfer settings, characterising failure modes and developing adaptive selection methods reducing catastrophic failures by 98.75%
Code: github.com/xrai keele/Classification-to-Segmentation

Data Analyst Intern - Explainable AI

2021-2022

Industry Partner via Keele University

- Developed explainable AI techniques for NLP models analysing sentiment in recruitment contexts
- Implemented interpretability methods ensuring transparency and fairness in automated decision-making systems

PUBLICATIONS

Journal Articles

Cockayne, M. J., Ortolani, M., Al-Bander, B. (2025). "DermFormer: Nested Multi-modal Vision Transformers for Robust Skin Cancer Detection." *Pattern Analysis and Applications*, Springer Nature, Volume 28, Article 194. DOI: [10.1007/s10044-025-01572-0](https://doi.org/10.1007/s10044-025-01572-0)

Cockayne, M. J., Ortolani, M., Al-Bander, B. “Sparse Concept Absence Network for Concept Bottleneck Models.” *Image and Vision Computing*, Elsevier. Status: Submitted.

Al-Bander, B., Ortolani, M., Dafaalla, M., **Cockayne, M. J.**, Pergola, G., Mamas, M. A. “Investigating and Mitigating Bias in Cardiovascular Disease Mortality Predictors for Scaling-up AI Fairness.” *Discover Computing*, Springer Nature. Status: Under Review.

Guo, S., **Cockayne, M. J.**, Ortolani, M., Al-Hamd, R., Al-Bander, B. “Zero-Shot Crack Segmentation for Structural Health Monitoring Using Class Activation Map Guided Segment Anything Model.” *Machine Vision and Applications*. Status: Under Review.

Conference Papers

Cockayne, M. J., Ortolani, M., Al-Bander, B. (2025). “Classification-to-Segmentation: Class Activation Mapping for Zero-Shot Skin Lesion Segmentation.” *AIHH 2025*, University of Cambridge. [Oral Presentation] DOI: [10.1007/978-3-032-00656-1_24](https://doi.org/10.1007/978-3-032-00656-1_24)

Cockayne, M. J., Ortolani, M., Al-Bander, B. “Fair Curriculum Learning for Concept Bottleneck Models in Dermatology.” *MICCAI 2026*. Status: Submitted.

TECHNICAL SKILLS

Programming: Python, LaTeX, R, SQL

Safe & Responsible AI: Fairlearn, Aequitas, adversarial debiasing, Fairness benchmarking, robustness benchmarking, uncertainty quantification

Explainable AI: CAM, GradCAM, integrated gradients, SHAP, LIME, attention visualisation, concept bottleneck models

ML Frameworks: PyTorch, TensorFlow, Keras, scikit-learn, Hugging Face

Foundation Models: Segment Anything Model (SAM/MedSAM), zero-shot transfer evaluation, failure mode analysis

Computer Vision: OpenCV, torchvision, TIMM

Data Analysis: Pandas, NumPy, SciPy, statistical modelling, information-theoretic analysis

Development: Git/GitHub, Jupyter Notebooks, Weights & Biases, TensorBoard

High Performance Computing: HPC cluster management, SLURM job scheduling, batch script optimisation

TEACHING EXPERIENCE

Laboratory Demonstrator

2022-Present

Keele University, School of Computer Science and Mathematics

- Supported undergraduate and master’s students across computer science modules for 4 years
- Delivered guest lectures on deep learning, computer vision, and responsible AI
- Mentored students on machine learning and AI ethics research projects

PROFESSIONAL SERVICE

Programme Committee: International Conference on AI in Healthcare (AIiH) 2026

Peer Review: Reviewer, 1st International Workshop on AI Safety and Security (AI-SS) 2026; *Engineering Applications of Artificial Intelligence* journal (2024-2025)

Leadership: Postgraduate Research Representative, Keele University (2022-2025, 3 years); Student Representative, Physics Programme (2018-2021, 3 years)

Mentorship: Supervised undergraduate and master’s research projects on machine learning and AI ethics; provided guidance on responsible AI methodologies and fairness evaluation techniques

Outreach: Data Science Content Creator Internship (2021-2022) - developed educational resources and materials on AI and data science fundamentals for broader audiences

Community: Turing Network Development Research Associate (2021-2022) - organised networking events, hackathons, and interdisciplinary research initiatives to foster cross-disciplinary collaboration

PRESENTATIONS AND TALKS

Oral Presentation: “Classification-to-Segmentation: Class Activation Mapping for Zero-Shot Skin Lesion Segmentation.” AIHH 2025, University of Cambridge (Sep 2025)

Research Seminars: Presented research on safe and responsible AI, robustness evaluation, fairness-aware learning, and interpretable architectures at departmental research seminars, Keele University (2022-2025)

Guest Lectures: Deep learning, safe and responsible AI (fairness, interpretability, bias mitigation), robustness evaluation, transformers in high-stakes domains (2022-2025)

REFERENCES

Available upon request